

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456823.8433 +/- 0.0034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.009 +/- 0.047
Transit depth (Rp/Rs)^2	0.0077 +/- 0.081 [%]
Orbital Inclination (inc)	82.68 +/- 2.89
Airmass coefficient 1 (a1)	0.9 +/- 0.49
Airmass coefficient 2 (a2)	0.17 +/- 0.35
Scatter in the residuals of the lightcurve fit is	61.71 %
Best Comparison Star	None
Optimal Aperture	0.75
Optimal Annulus	3.75
Transit Duration (day)	0.028 +/- 0.032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.009 ± 0.047 vs 0.1326 (deviation 2.62σ)
Duration fit vs published	0.028 d vs 0.0737 d (deviation 1.43σ)
Mid-time O–C	5.5 min
Depth SNR	0.2
Residual scatter	61.71 %

Light curve

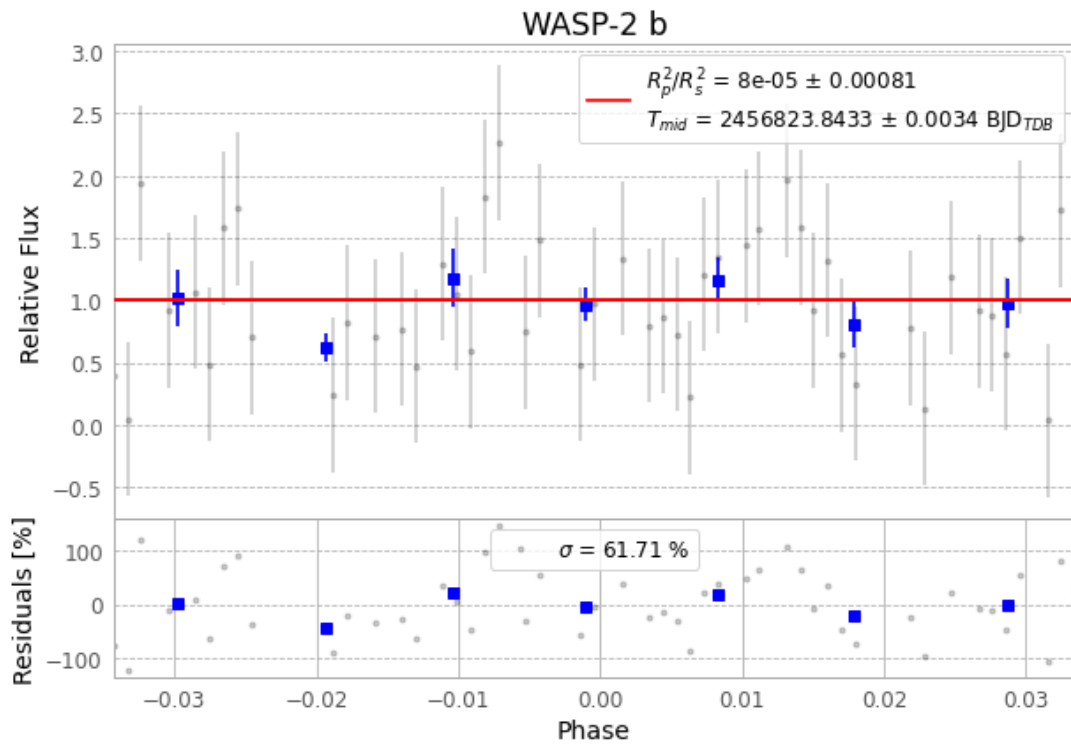


Figure 1 — FinalLightCurve_WASP-2 b_2014-06-14.png (SHA-256 217e5344c0c8501d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [106, 460], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ebcf533e9d01922c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

74 FITS frames (49 MB), WASP-2140615062112.FITS → WASP-2140615100012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-140615.log (a290fd636f4c...), FinalLightCurve_WASP-2 b_2014-06-14.png (217e5344c0c8...), FinalLightCurve_WASP-2 b_2014-06-14.csv (0277da433d95...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 05:54Z.